MODULE 3: MACHINE TOOLS, AUTOMATION & ROBOTICS

Machining: Machining is the process of removing the excess material from the work piece by using a Machine tool

Machine Tool: Machine Tool is a Power driven machine used to perform Machining operations

Examples of Machine Tools

1. Lathe Machine
2. Drilling Machine
3. Milling Machine
4. Grinding Machine
5. Shaping Machine

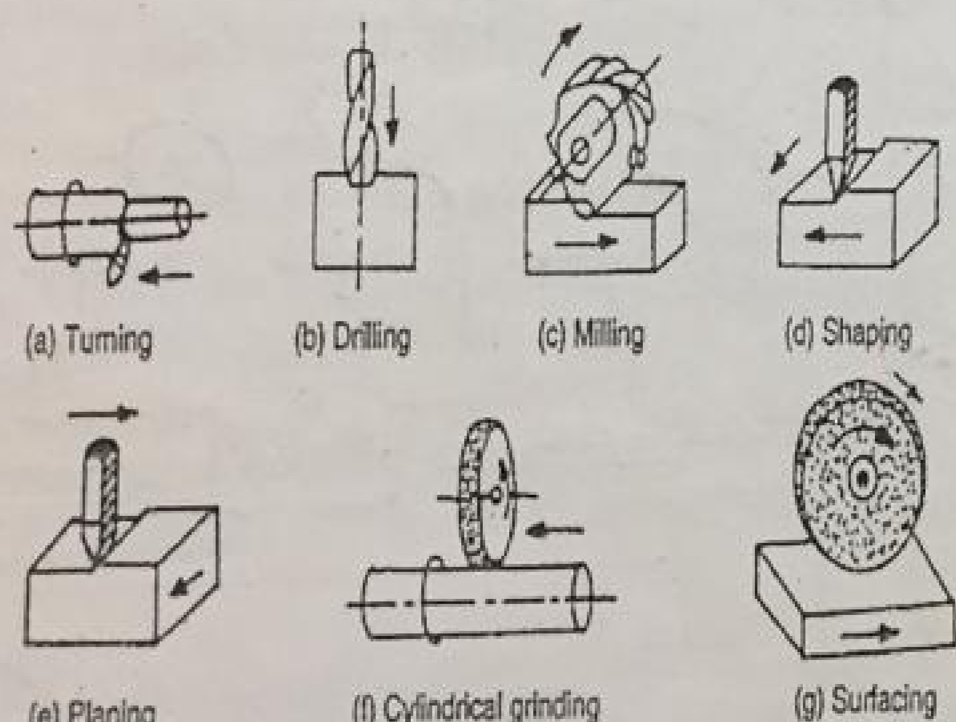


Fig: Different types of Machining Operations

LATHE:

- Lathe is defined as A Machine Tool, used to remove excess material from the work piece by forcing a cutting tool against the rotating work piece
- Lathe is also called as Turning Machine since the work piece is turned or rotated between the two centers.
- Lathe is known as **Mother of Machine Tools.**

Classification of Lathes:

a) Speed Lathe	d) Tool room Lathe
b) Engine Lathe	e) Capstan and Turret Lathe
c) Bench Lathe	f) Automatic Lathe

Working Principle of Lathe:

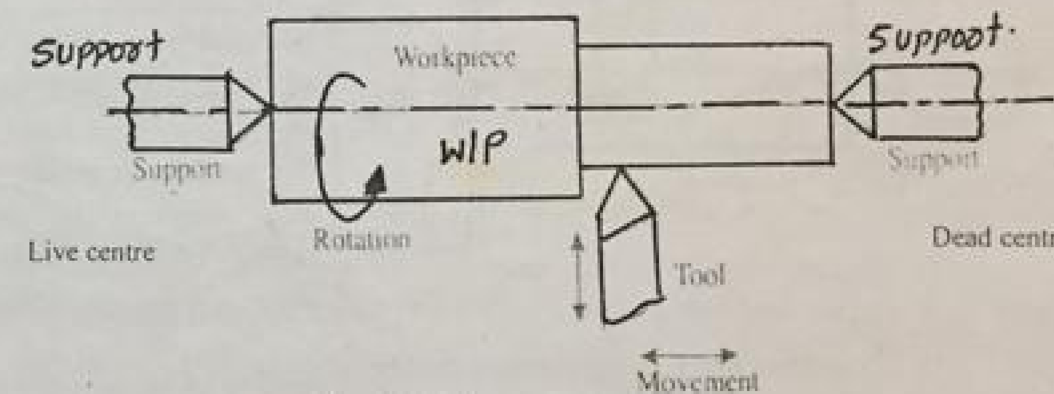


Fig: Working principle of Lathe

- Fig shows the working principle of Lathe
- The work piece or job is held between the two supports(Live centre and dead centre)
- The cutting tool is held in the Tool post
- The Tool can be moved in Longitudinal and Transverse direction
- The tool material should be harder than work piece material
- The Lathe works on the principle that, a cutting tool removes excess material when it is moved or forced against a rotating work piece.

MAJOR PARTS OF LATHE (CENTRE LATHE)

1. BED
2. HEADSTOCK
3. TAILSTOCK
4. CARRIAGE ASSEMBLY
5. FEED ROD
6. LEAD SCREW

1. Bed:

- The bed is the base of the lathe. It is made of cast iron.
- Function of bed is to support all other elements of the lathe.

2. Headstock :

- It is permanently fastened to the left hand end of the lathe bed.
- Function is to support spindle & house the main drive.
- Spindle is a hollow rotating shaft used for holding workpiece.
- Generally chuck is mounted on the spindle.
- The main drive is used to drive spindle at different speeds.

3. Tailstock :

- The tailstock is located at the right hand end of the lathe bed.
- It can be moved along the guideways on the lathe bed & clamped in any position along lathe bed.
    1. It's function is to support the long workpieces during machining.
    2. Also to hold tools like drill, reamer or tap.

4. Carriage :

- Located between the headstock & the tailstock.
- It slides along the guideways on the lathe bed.
- It's function is to hold the cutting tool & also to give longitudinal & cross feed to the cutting tool.

5. Lead Screw :

- It's the long threaded shaft driven by feed drive.
- It's function is to give mechanized motion to carriage for cutting threads on the workpiece.

LATHE OPERATIONS:

The following operations are performed on the lathe machine

1. Turning operation
2. Facing operation
3. Knurling operation
4. Thread cutting operation
5. Taper turning operation

1. Turning Operation:

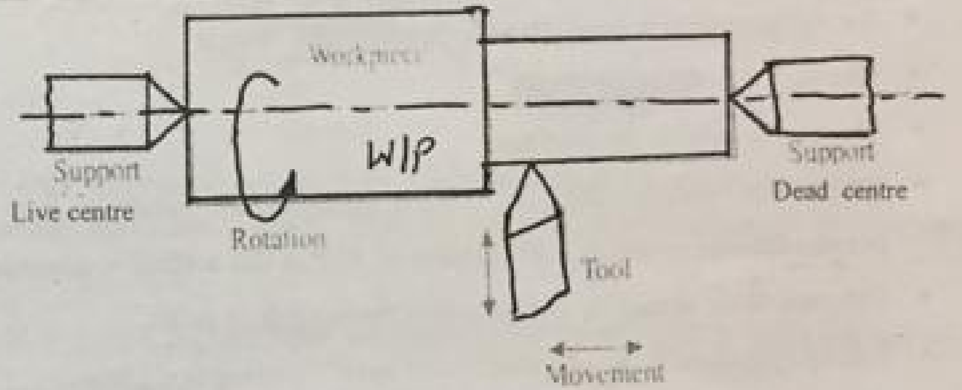
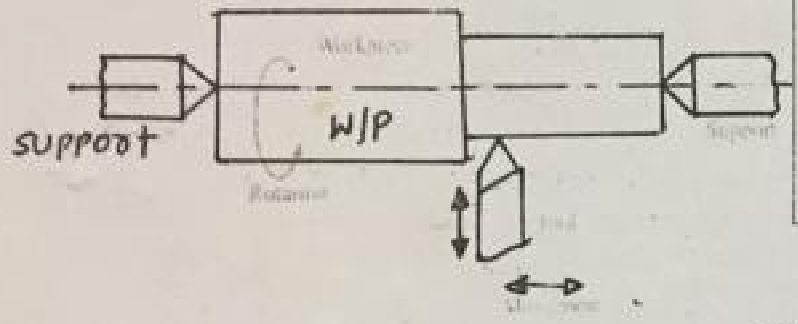


Fig: Turning operation

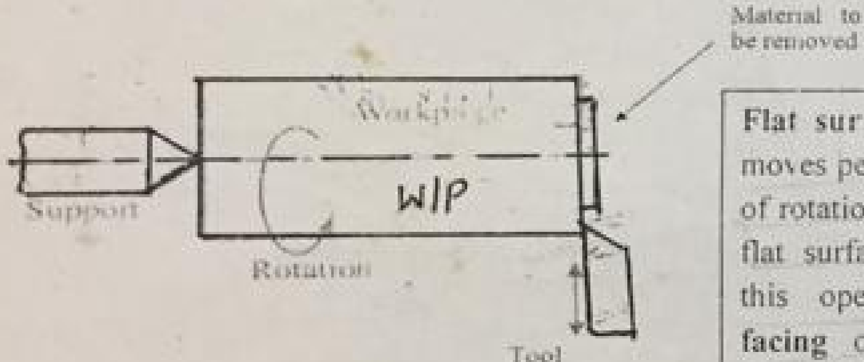
1 - Fig shows the Turning operation
2 - Turning operation is performed to produce cylindrical surface and hence this operation is also called as **cylindrical** turning or plain turning
3 - The W/P is held between the Live centre and dead centre
4 - The tool is held in the tool post
5 - When the cutting tool is moved against the rotating W/P and moved parallel to the axis of rotation of W/P, a cylindrical surface is generated
6 - This operation is carried out to reduce the diameter of W/P

Basically, three types of surface can be produced depending on the movement of the cutting tool

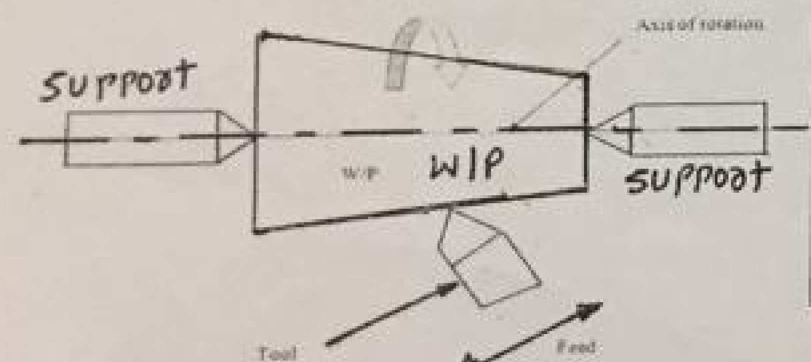
1. *Cylindrical surface:*
2. *Flat surface:*
3. *Tapered surface:*



Cylindrical surface: When the tool moves parallel to the axis of rotation of the work piece, a cylindrical surface is produced. And this operation is called as **turning operation.** The figure shows the Turning operation



Flat surface: When the tool moves perpendicular to the axis of rotation of the work piece, a flat surface is produced. And this operation is called as **facing** operation. The figure shows the Facing operation.



Tapered surface: When the tool moves at an angle to the axis of rotation of the work piece, a tapered surface is produced. And this operation is called as Taper turning operation. The figure shows the

PARTS OF LATHE (CENTRE LATHE):

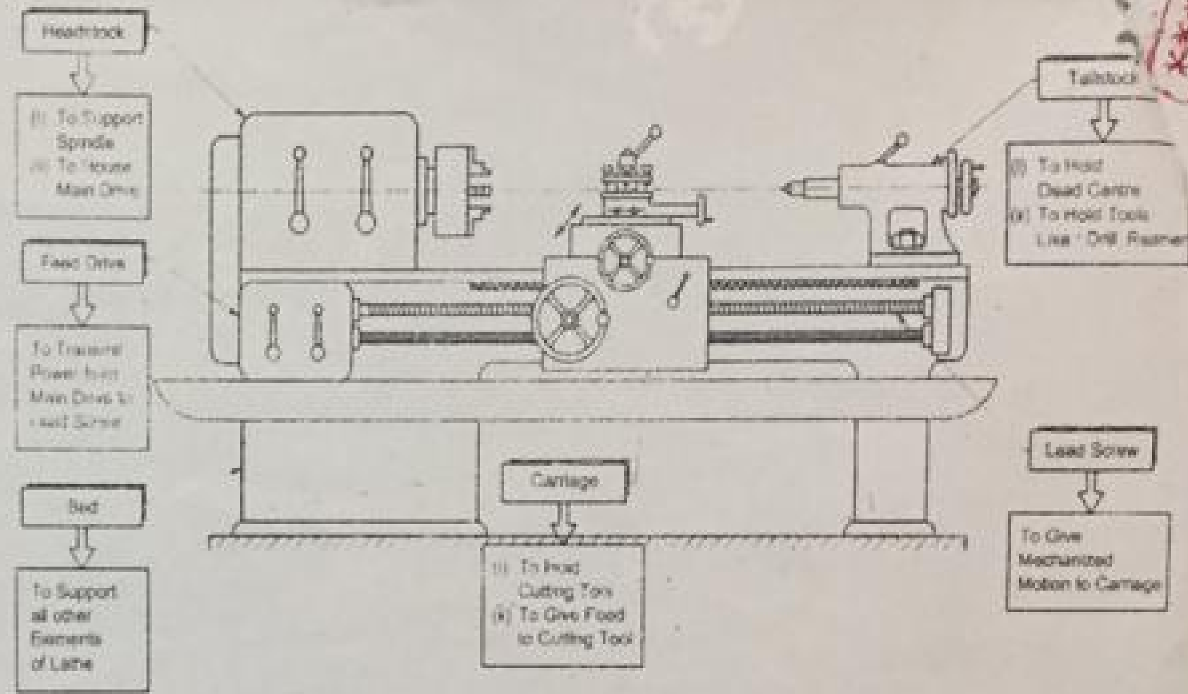


Fig : Parts of center Lathe

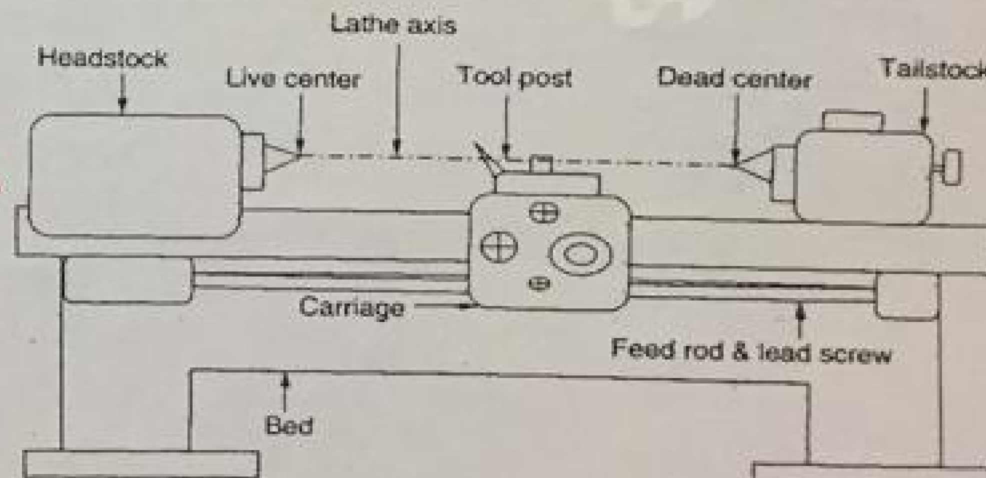
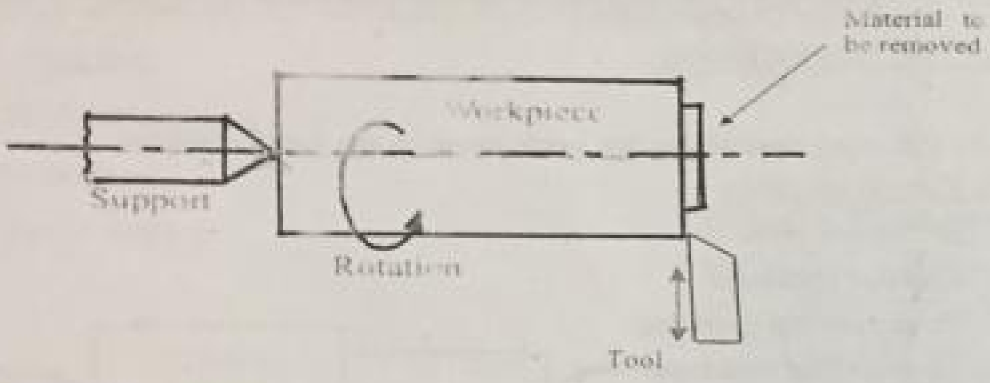


Fig: Line diagram of Lathe machine

2. Facing Operation:



- Fig shows the Facing operation
- Facing operation is performed to produce flat surface
- The W/P is held in the chuck
- The tool is held in the tool post
- When the cutting tool is moved against the rotating W/P and moved **perpendicular** to the axis of rotation of W/P, a flat surface is generated
- This operation is carried out to reduce the **length** of W/P
- Facing operation is usually performed before turning operations.

3. Knurling Operation:

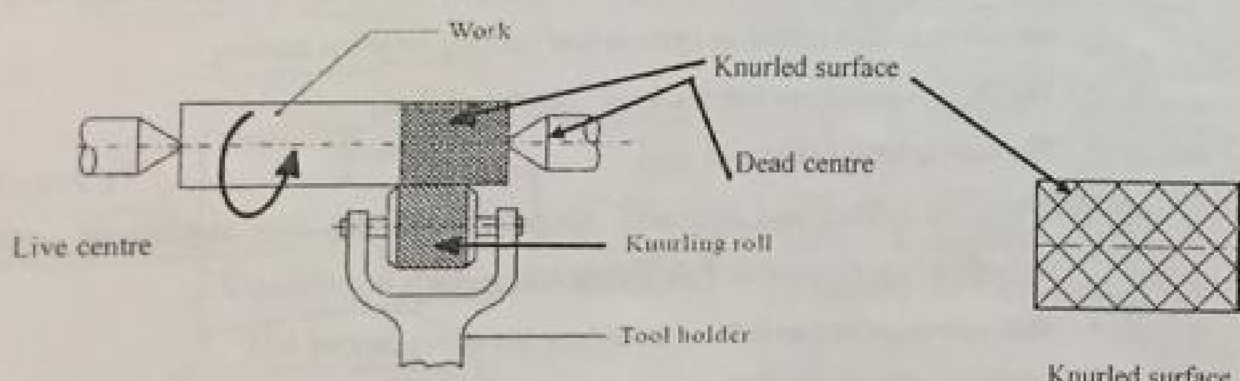


Fig: Knurling operation

- Fig shows Knurling operation
- Knurling is the operation of producing a diamond shaped pattern on the surfa W/P for gripping purpose
- Knurling operation is carried out using a Knurling tool
- The Knurling tool is held rigidly on the tool post
- In the operation, the Knurling tool is pressed against the rotating the W/P until the tool produces a diamond shaped pattern on the W/P

4. Thread Cutting Operation :

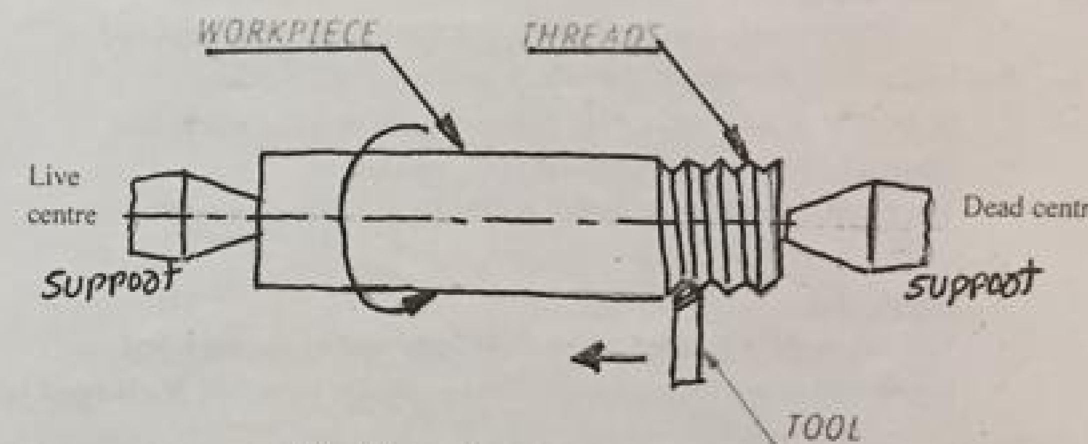


Fig: Thread cutting operation

- The fig shows Thread cutting operation
- The W/P is held between the chuck and dead centre
- The thread cutting tool is held in the tool post
- The principle of thread cutting is to produce a helical groove on a cylindrical or conical surface
- The shape of tool resembles (matches) the thread profile
  - ➢ To cut V-thread profile, pointed tool is used
  - ➢ To cut square thread profile, square end tool is used

5. Taper Turning operation:

- Taper turning is the operation of producing a conical surface from a cylindrical surface
- Taper surface is produced, when the tool moves at an angle to the axis of rotation of W/P

Taper Elements:

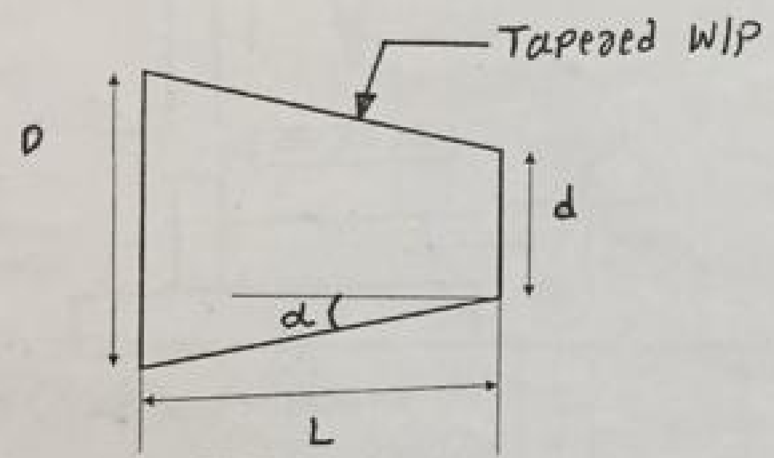


Fig Taper Surface

Where
D= large diameter of taper
d= small diameter of taper
L= length of taper
α = half taper angle

$$\boxed{\text{Therefore, Taper } = \frac{D-d}{2L}}$$

Taper Turning Methods:

In the Lathe, Taper surface can be produced by following methods

1. **Taper turning By Swivelling(rotating) the Compound rest**
2. **Taper turning By Tailstock set over method or Tailstock Off-set method:**

MACHINE TOOLS, AUTOMATION & ROBOTICS

1. Taper turning By Swivelling(rotating) the Compound rest:

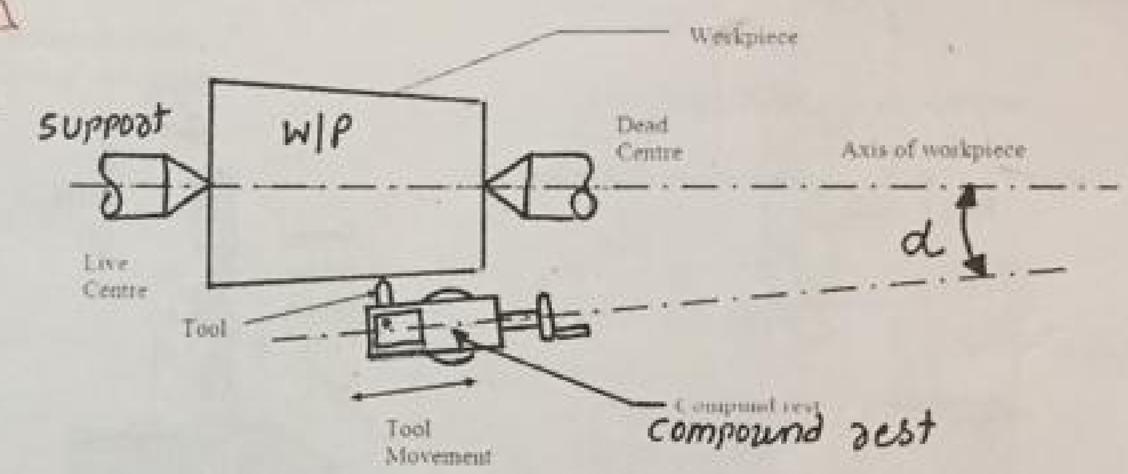


Fig: Taper turning By Swivelling(rotating) the Compound rest

- Fig shows the Taper Turning operation by Swivelling(rotating) the Compound rest
- The W/P is held Between live and dead centre and the tool is mounted(placed) on the compound rest
- The tool mounted on the compound rest has a circular base which is graduated in degrees
- In this method, the compound rest is swivelled (rotated) at an angle 'α' to the lathe axis
- The angle 'α' is called as taper angle
- The taper angle is calculated by using formula Tan α = $\frac{D-d}{2L}$
- The tool in the compound rest is fed against the rotating W/P by compound rest hand wheel
- As the tool moves over the W/P, a taper surface is produced as shown in above figure

DRILLING MACHINES AND ITS OPERATIONS

DRILLING: Drilling is the operation of producing circular hole in the work-piece by using a rotating tool called as DRILL BIT or Twist drill.

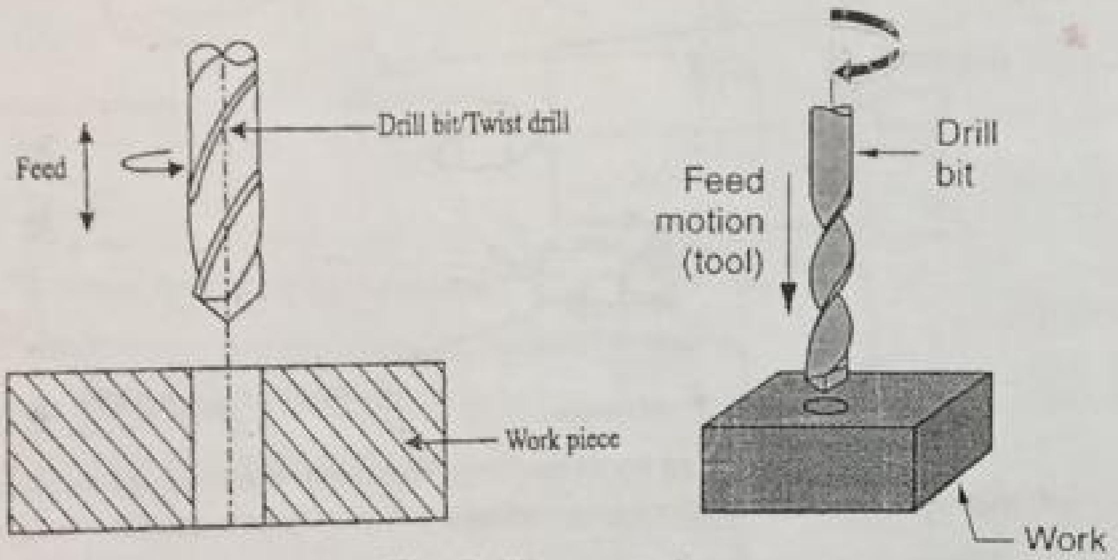


Fig: Drilling operation

- The drill bit has sharp conical edge at the bottom and twisted edges around its body with Helical groove
- The machine used for drilling is called **Drilling Machine.**
- In drilling machine, the work piece is held stationary ie. Clamped in position and the drill bit rotates to make a hole.
- The drilling operation can also be accomplished in lathe, in which the drill bit is held in tailstock and the work piece is held in the chuck.

Classification of Drilling Machines

1. Portable drilling machine
2. Bench drilling machine or Sensitive drilling machine
3. Radial drilling machine
4. Multi-spindle drilling machine
5. Automatic drilling machine

1. SENSITIVE OR BENCH DRILLING MACHINE:

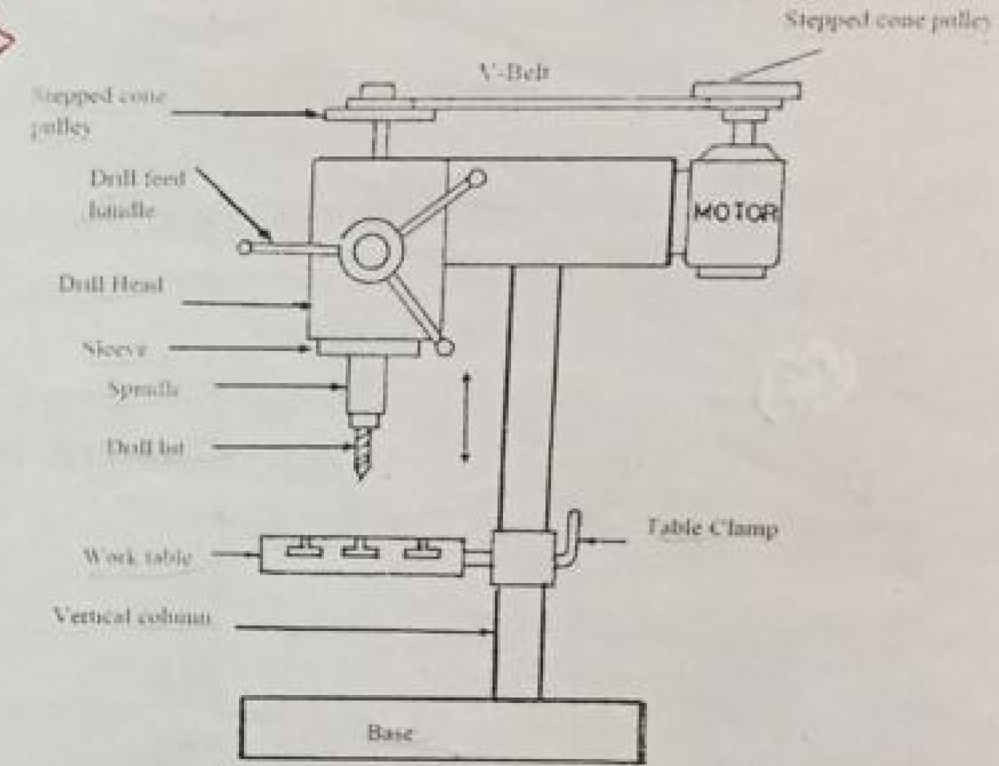


Fig Sensitive or Bench Drilling Machine

- Fig shows Sensitive or Bench Drilling Machine
- Sensitive or Bench Drilling Machine is used for producing or drilling small holes upto 15mm
- This machine is generally small sized and it is mounted on a work-bench

Components of Bench drilling machine

1. Base
2. Vertical column
3. Work table
4. Drill head

DRILLING OPERATIONS

Operations that can be performed in a drilling machine are

1. Drilling Operation
2. Boring Operation
3. Reaming Operation
4. Tapping Operation
5. Countersinking Operation
6. Counter Boring Operation

1. Drilling Operation:

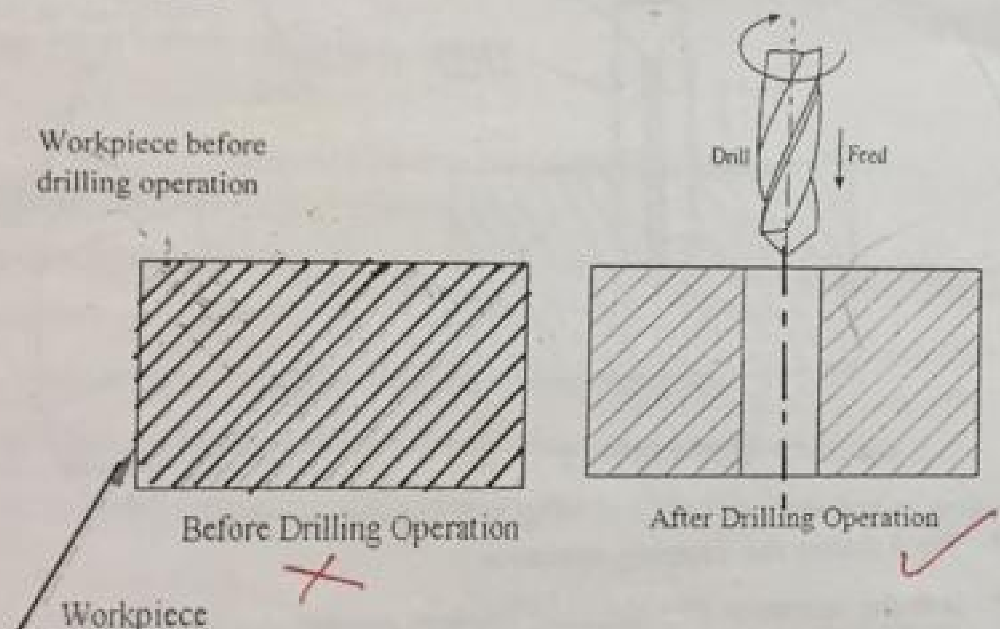
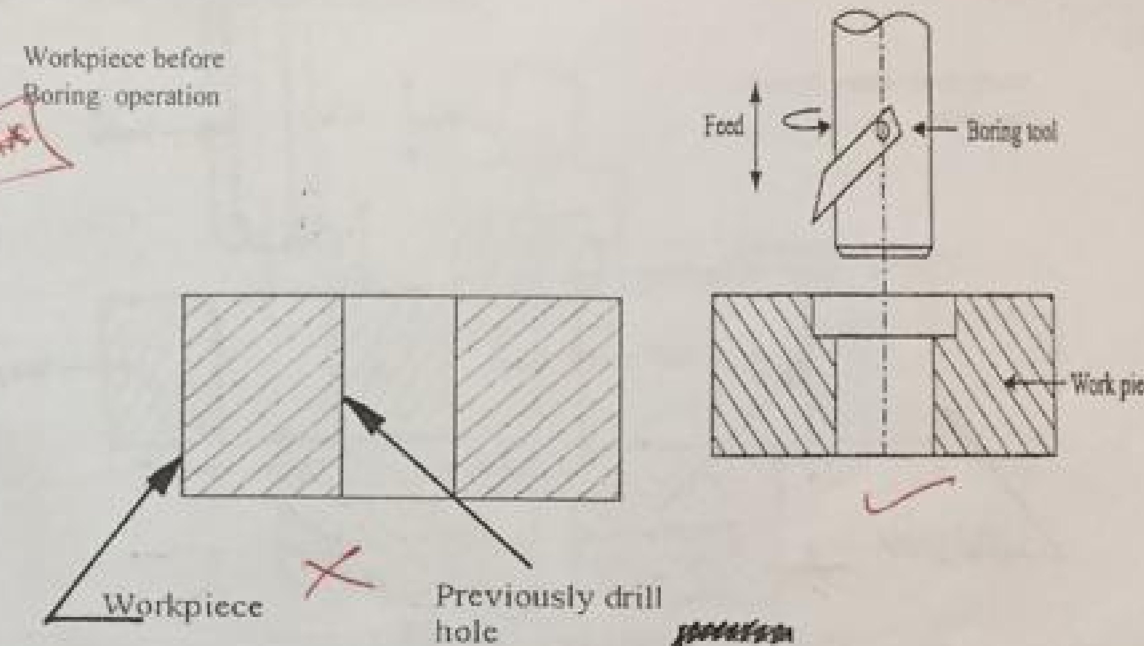


Fig 1: drilling operation

1 ▪ Fig 1 shows the drilling operation
2 ▪ It is the operation of producing a circular hole in a solid W/P
3 ▪ It is carried out by drill bit
4 ▪ The W/P is rigidly clamped on the work table
5 ▪ When the revolving drill bit is forced into the W/P, a circular hole is generated

2. Boring Operation:



- Fig 2 shows the Boring operation
- It is the operation of enlarging a previously drilled hole
- It is carried out by boring tool which has only one cutting edge
- The boring tool is held by a boring bar
- Boring operation is performed at very low speed

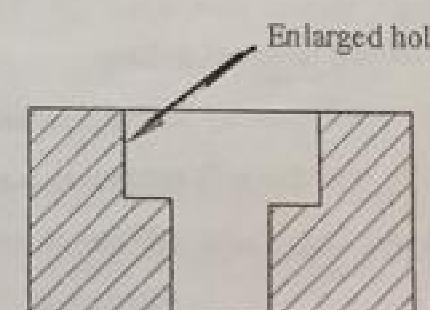


Fig: Enlarged hole after Boring operation

3. Reaming operation:

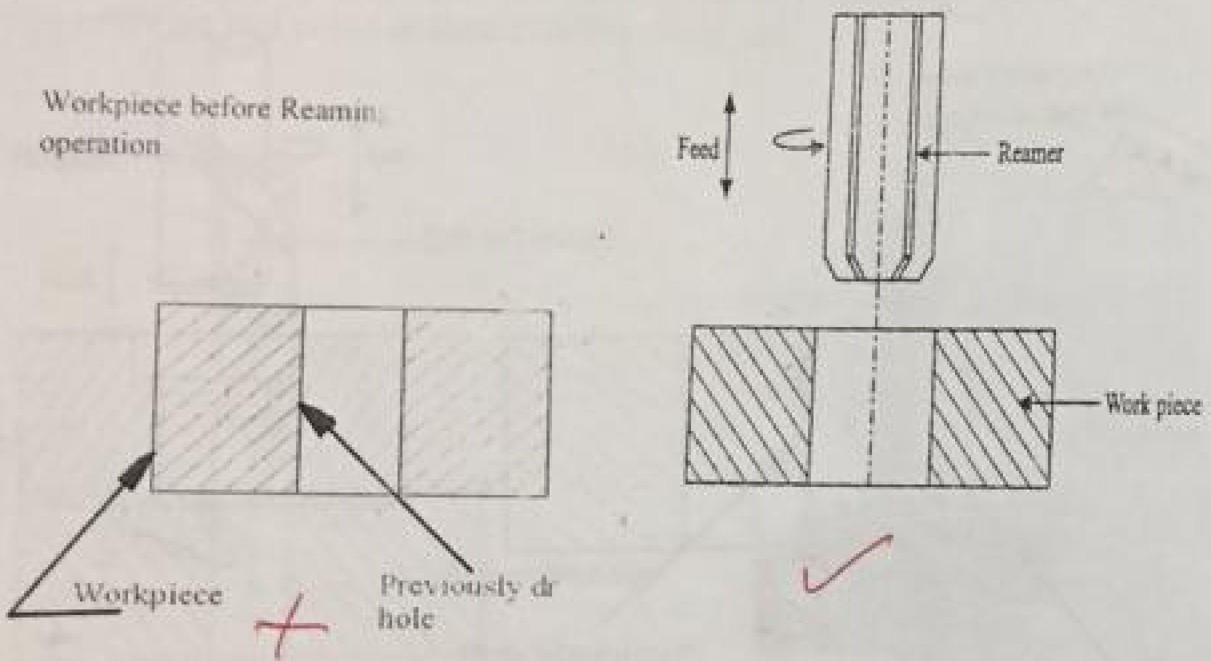


Fig 3: Reaming operation

- Fig 3 shows the Reaming operation
- It is the operation of producing a smooth surface finish in a previously drilled hole and to bring it to the accurate size
- It is carried out by a cutting tool called as 'Reamer'. Reamer has multiple edges around its periphery
- Reaming is done at the half speed of drilling
- In this operation, very small quantity of metal is removed
- The surface obtained by reaming will be smoother and the size accurate

4. Tapping Operation:

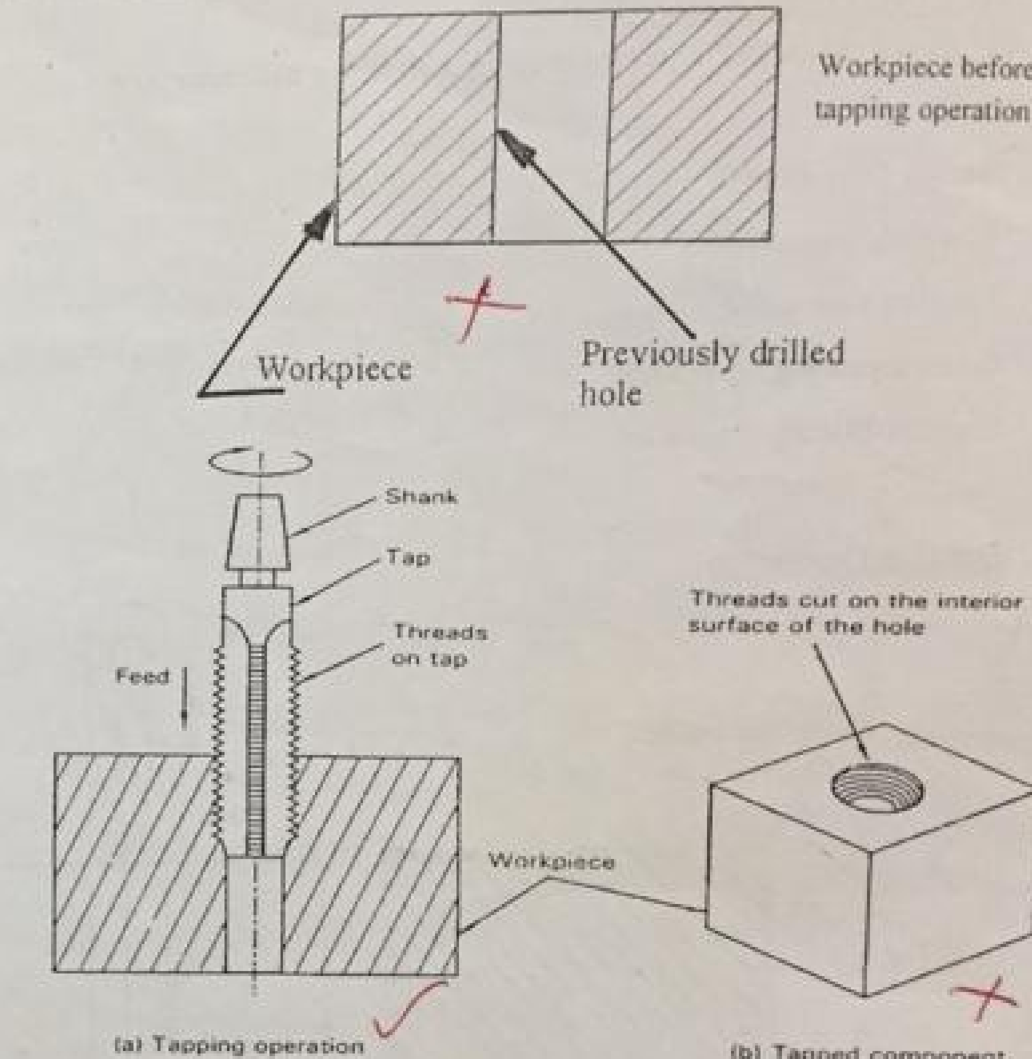


Fig 4: Tapping operation

- Fig 4 shows the Tapping operation
- It is the operation of producing internal threads in a previously drilled hole
- It is carried out by a cutting tool called as 'Tap'
- Tap has cutting edges in the shape of threads
- When the Tap is rotated in the W/P, it removes the metal and hence prod internal threads in the W/P

5. Counter Sinking Operation:

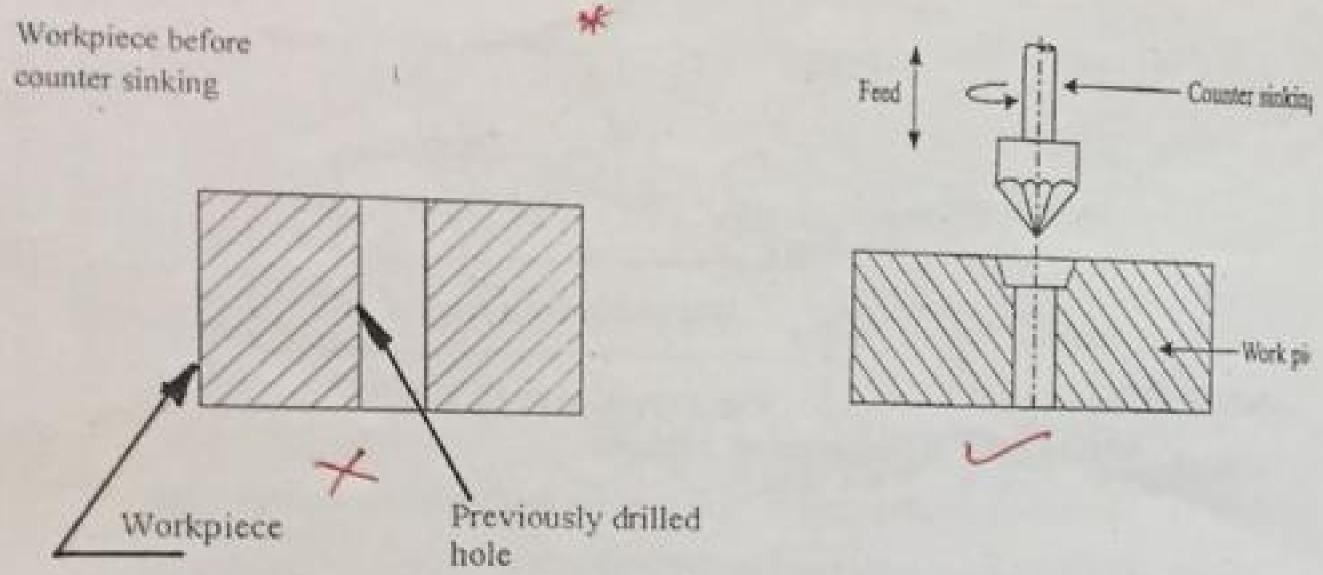
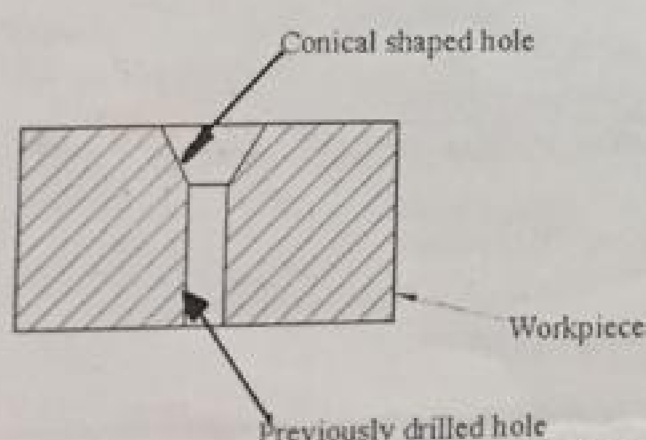


Fig 5: Counter sinking operation

- Fig 5 shows the Counter sinking operation
- It is the operation of enlarging one end of a previously drilled into a conical shape for a small depth
- It is carried out by a cutting tool called as 'counter sink tool'
- The cutting edges of the tool are formed(made) at the conical surface
- The cutting speed for countersinking operation is 25% less than that of drilling.



6. Counter Boring Operation:

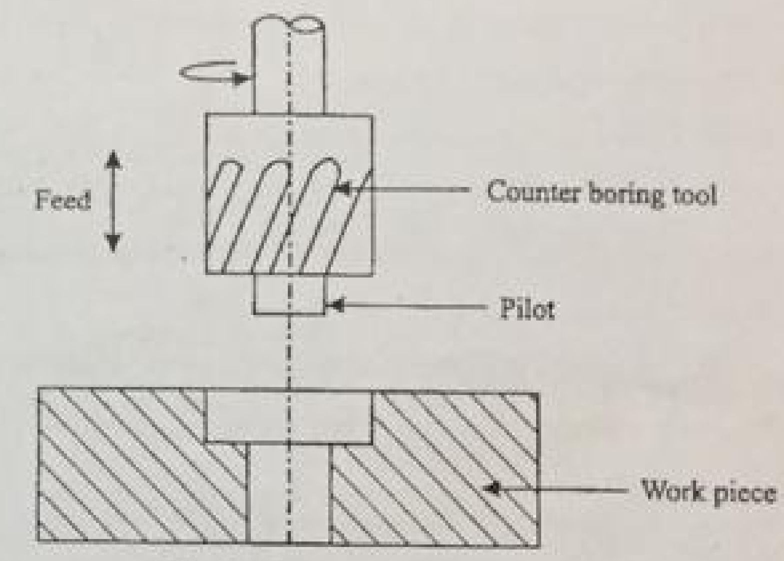


Fig 6: Counter boring operation

- Fig 6 shows the Counter boring operation
- It is the operation of enlarging one end of a previously drilled hole into a cylindrical shape (of diameter more than the previously drilled hole) for a small depth
- It is carried out by a cutting tool called as 'counter boring tool'
- The counter boring tool are made with cutting edges which may be straight or helical
- The tool is provided with a pilot
- The size of the pilot should be equal to the size of previously drilled hole
- During operation, pilot enters the previously drilled hole and maintains the alignment of the counter boring tool
- Counter boring holes (large sized holes) are useful for accommodating heads of bolts, studs etc.,